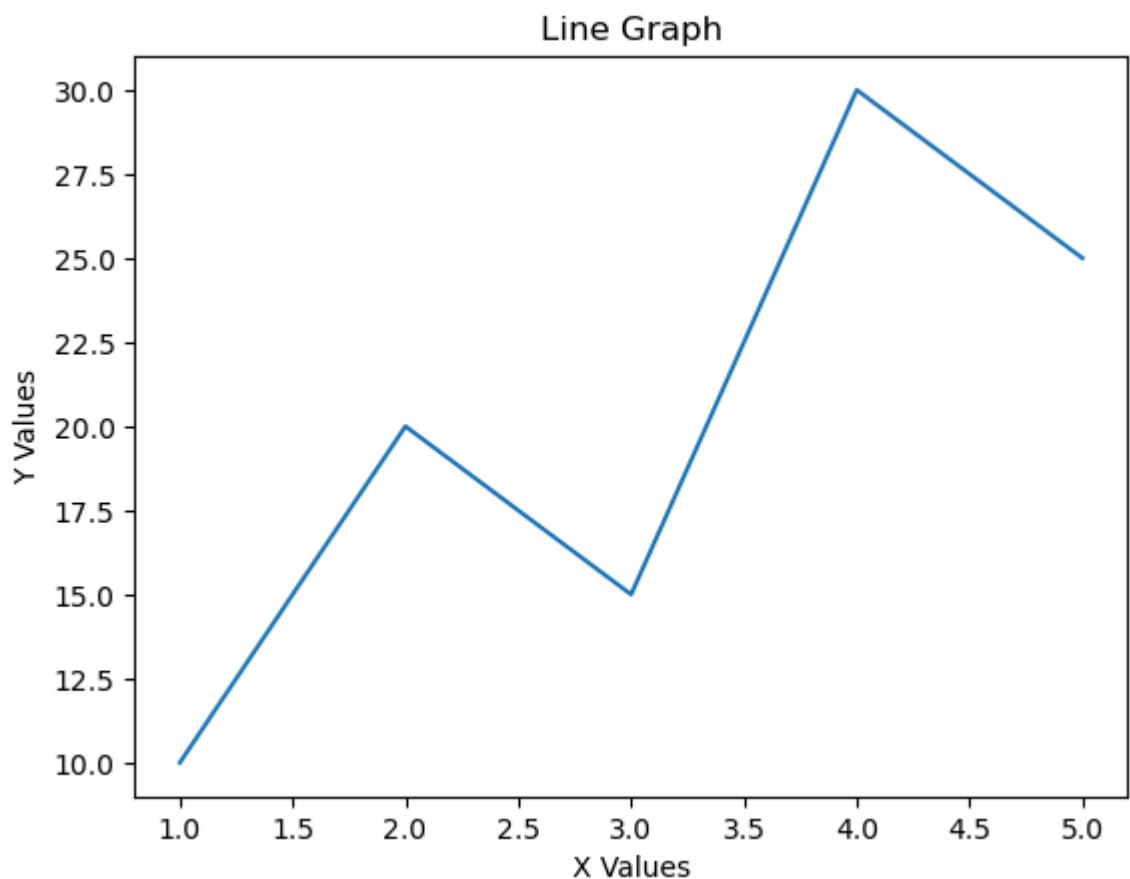


3.To visualize data using Matplotlib graphs on real datasets.

```
In [1]: #Importing Matplotlib  
  
import matplotlib.pyplot as plt
```

```
In [2]: #1. Line Graph  
  
import matplotlib.pyplot as plt  
  
x = [1, 2, 3, 4, 5]  
y = [10, 20, 15, 30, 25]  
  
plt.plot(x, y)  
plt.title("Line Graph")  
plt.xlabel("X Values")  
plt.ylabel("Y Values")  
plt.show()
```

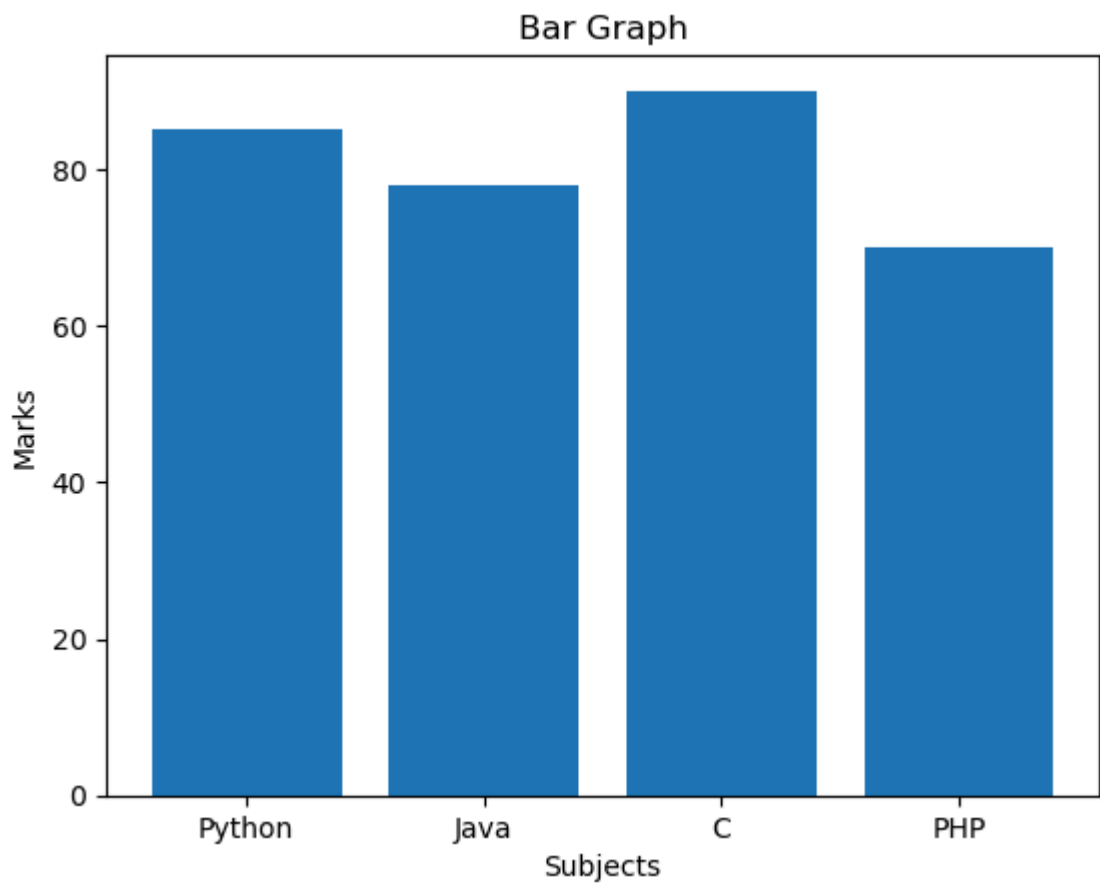


In [3]: #2. Bar Graph

```
import matplotlib.pyplot as plt

subjects = ['Python', 'Java', 'C', 'PHP']
marks = [85, 78, 90, 70]

plt.bar(subjects, marks)
plt.title("Bar Graph")
plt.xlabel("Subjects")
plt.ylabel("Marks")
plt.show()
```

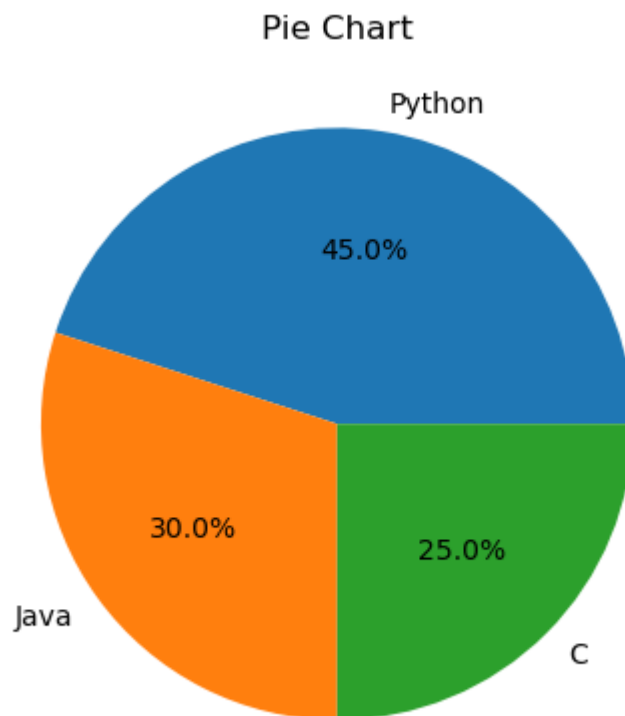


In [4]: #3. Pie Chart

```
import matplotlib.pyplot as plt

labels = ['Python', 'Java', 'C']
sizes = [45, 30, 25]

plt.pie(sizes, labels=labels, autopct='%1.1f%%')
plt.title("Pie Chart")
plt.show()
```

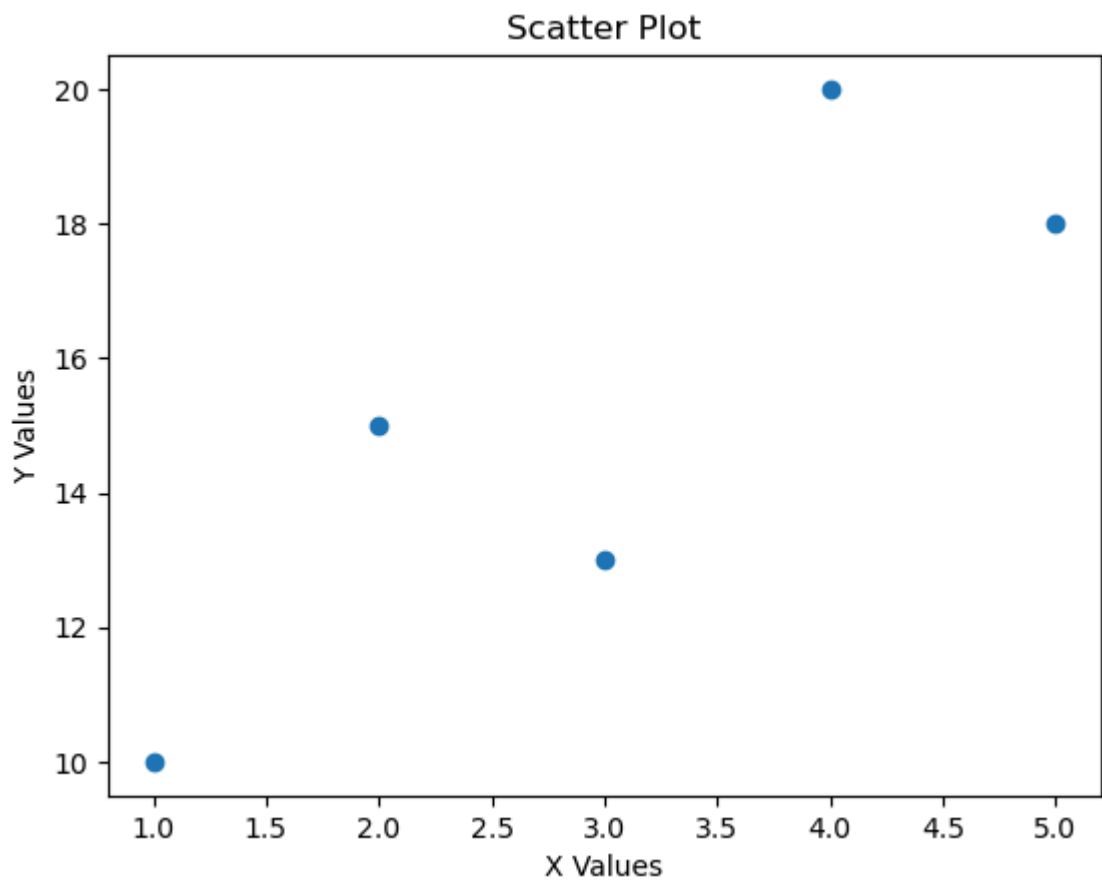


In [5]: #4. Scatter Plot

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [10, 15, 13, 20, 18]

plt.scatter(x, y)
plt.title("Scatter Plot")
plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.show()
```

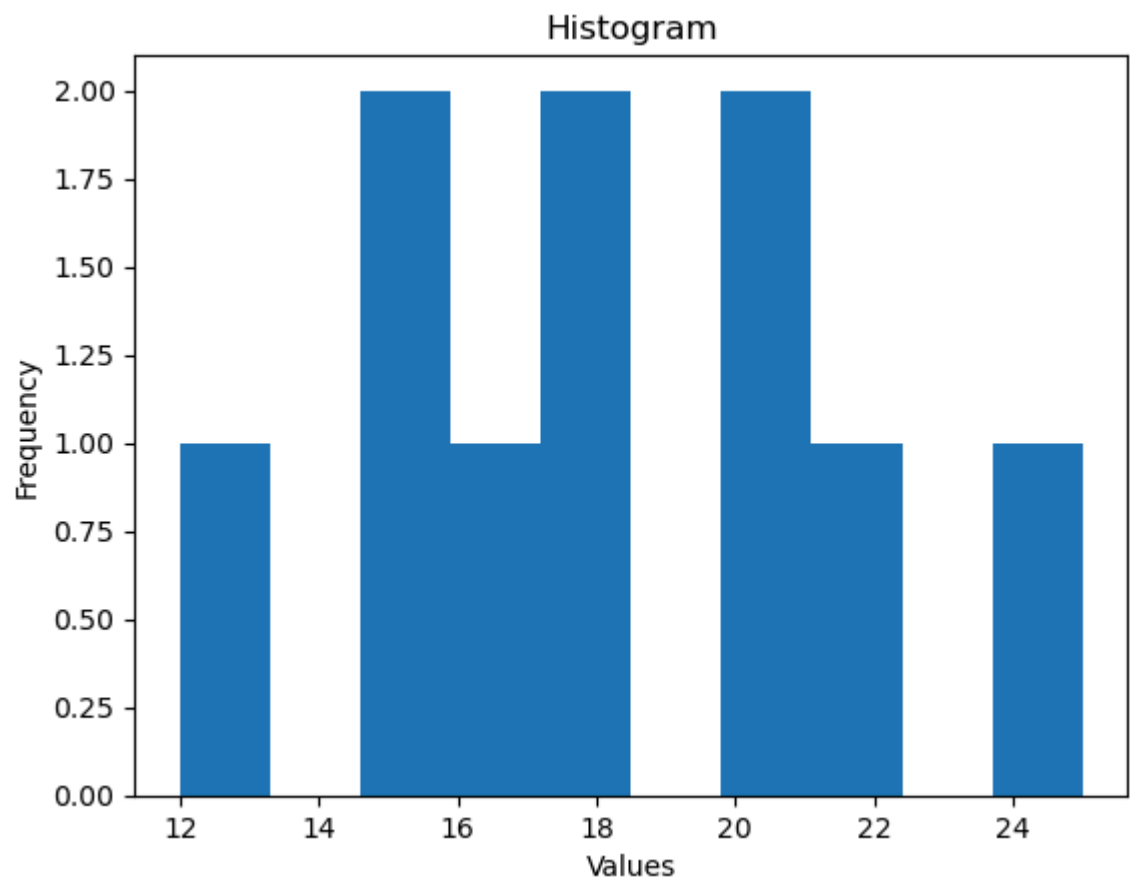


In [6]: *#5. Histogram*

```
import matplotlib.pyplot as plt

data = [12, 15, 18, 20, 22, 25, 18, 20, 15, 17]

plt.hist(data)
plt.title("Histogram")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.show()
```



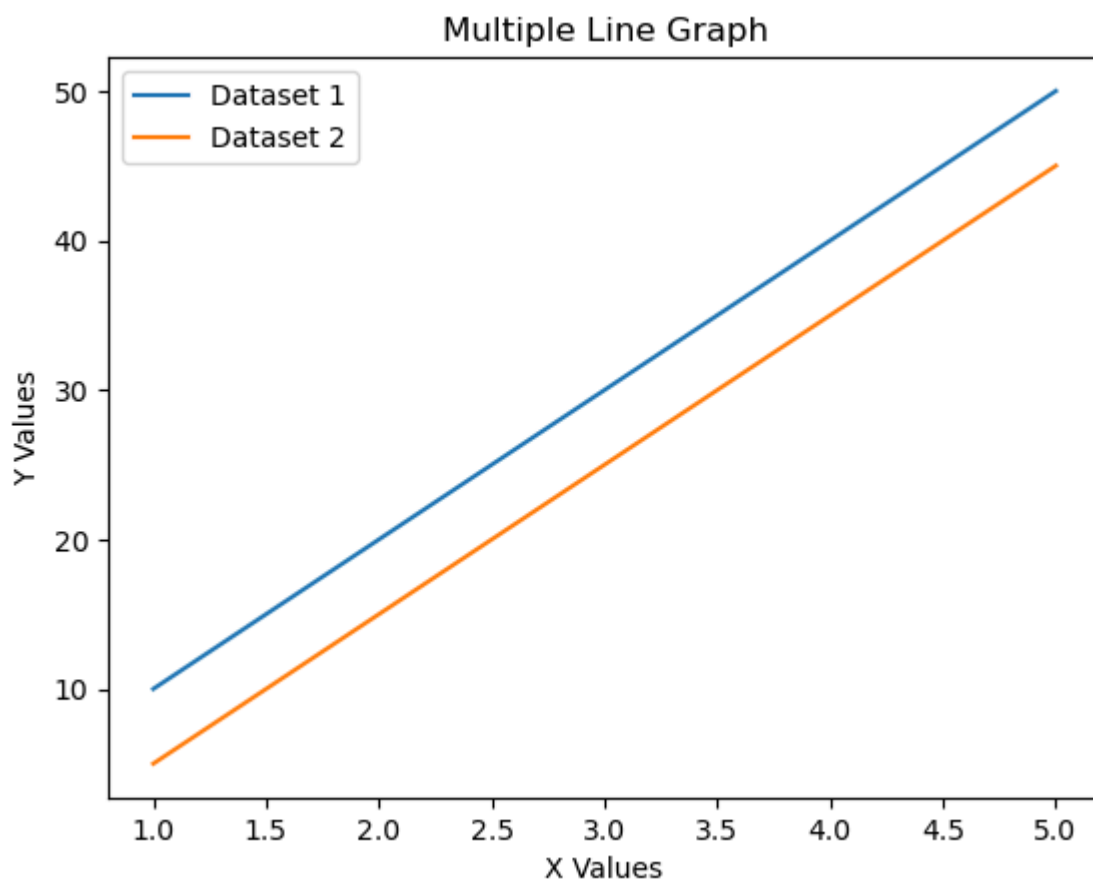
In [7]: *#6. Multiple Line Graph*

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y1 = [10, 20, 30, 40, 50]
y2 = [5, 15, 25, 35, 45]

plt.plot(x, y1, label='Dataset 1')
plt.plot(x, y2, label='Dataset 2')

plt.title("Multiple Line Graph")
plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.legend()
plt.show()
```



In [1]: *# Practical 4*

```
import pandas as pd
import matplotlib.pyplot as plt
```

In [3]: *#Load the Dataset*

```
df = pd.read_csv(r"C:\Users\Smit\OneDrive\Desktop\Dataset\netflix_tv_shows_det
```

```
In [4]: #Display the Dataset
print(df.head())
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	genres	language	description	popularity	vote_count	vote_average
0	33238	TV Show	Running Man	안재철	Yoo Jae-suk, Jee Seok-jin, Kim Jong-kook, Haha...	South Korea	2010-07-11	2010	8.241	1 Seasons	Comedy, Reality	ko	A reality and competition show where members a...	1929.898	187	8.241
1	32415	TV Show	Conan	NaN	Conan O'Brien, Andy Richter	United States of America	2010-11-08	2010	7.035	1 Seasons	Talk, Comedy, News	en	A late night television talk show hosted by C...	1670.580	229	7.035
2	37757	TV Show	MasterChef Greece	NaN	NaN	Greece	2010-10-03	2010	5.600	1 Seasons	Reality	el	MasterChef Greece is a Greek competitive cooki...	1317.092	6	5.600
3	75685	TV Show	Prostřeno!	NaN	Václav Vydra, Jana Boušková	Czech Republic	2010-03-01	2010	6.500	1 Seasons	Reality	cs	The knives (and forks) are out as a group of s...	1095.776	6	6.500
4	33847	TV Show	The Talk	NaN	Amanda Kloots, Jerry O'Connell, Akbar Gbaja-Bi...	United States of America, Ireland	2010-10-18	2010	3.400	1 Seasons	Talk	en	A panel of well-known news and entertainment p...	712.070	12	3.400

```
In [5]: #Display Column Names
print(df.columns)
```

```
Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_adde
d',
      'release_year', 'rating', 'duration', 'genres', 'language',
      'description', 'popularity', 'vote_count', 'vote_average'],
      dtype='object')
```

```
In [6]: #Dataset Information
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 16000 entries, 0 to 15999
Data columns (total 16 columns):
#   Column                Non-Null Count  Dtype
---  -
0   show_id                16000 non-null  int64
1   type                  16000 non-null  object
2   title                 16000 non-null  object
3   director              5035 non-null   object
4   cast                  14843 non-null  object
5   country               14203 non-null  object
6   date_added            16000 non-null  object
7   release_year          16000 non-null  int64
8   rating                16000 non-null  float64
9   duration              16000 non-null  object
10  genres                 15026 non-null  object
11  language               16000 non-null  object
12  description            12794 non-null  object
13  popularity             16000 non-null  float64
14  vote_count             16000 non-null  int64
15  vote_average           16000 non-null  float64
dtypes: float64(3), int64(3), object(10)
memory usage: 2.0+ MB
None
```

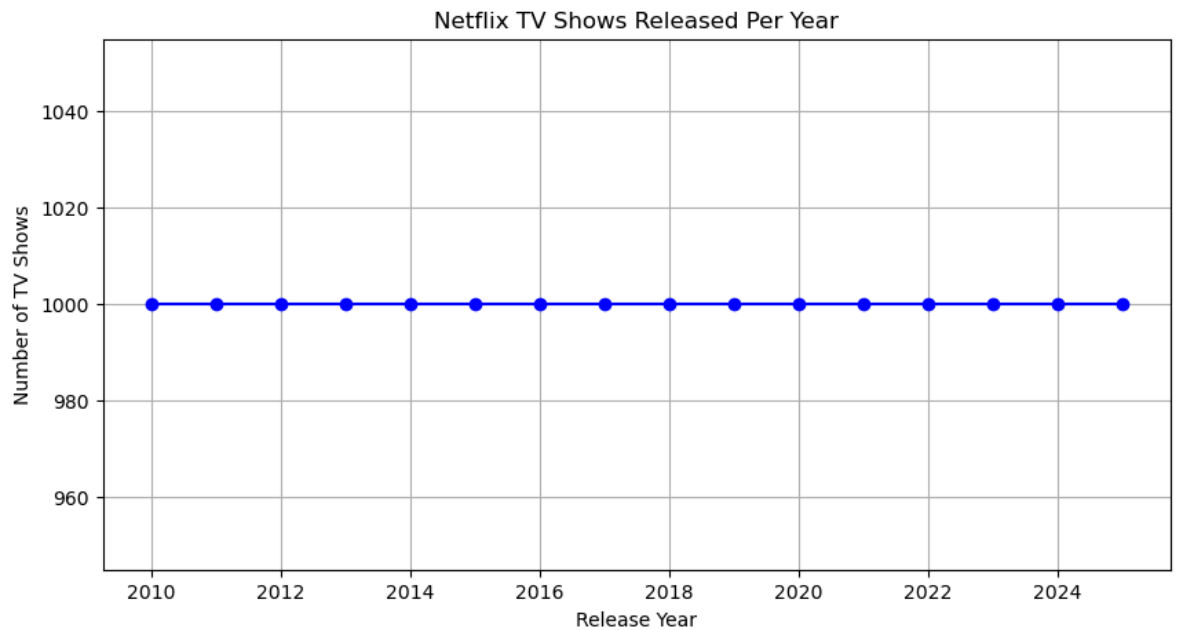


```
In [7]: #Line Graph (TV Shows Released Per Year)
release_year = df['release_year'].value_counts().sort_index()

plt.figure(figsize=(10,5))
plt.plot(release_year.index, release_year.values,
         marker='o', color='blue')

plt.title("Netflix TV Shows Released Per Year")
plt.xlabel("Release Year")
plt.ylabel("Number of TV Shows")
plt.grid(True)

plt.show()
```

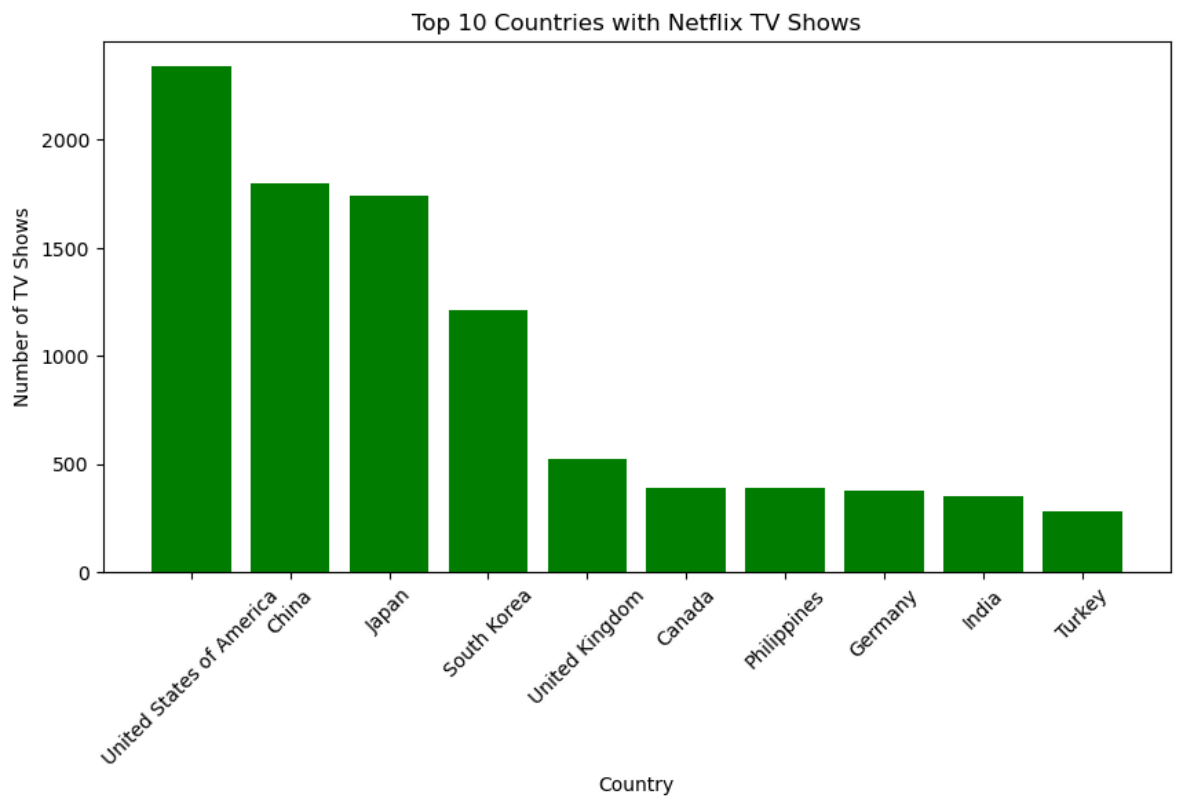


```
In [8]: #Bar Graph (Top 10 Countries)
top_country = df['country'].value_counts().head(10)

plt.figure(figsize=(10,5))
plt.bar(top_country.index, top_country.values,
        color='green')

plt.title("Top 10 Countries with Netflix TV Shows")
plt.xlabel("Country")
plt.ylabel("Number of TV Shows")
plt.xticks(rotation=45)

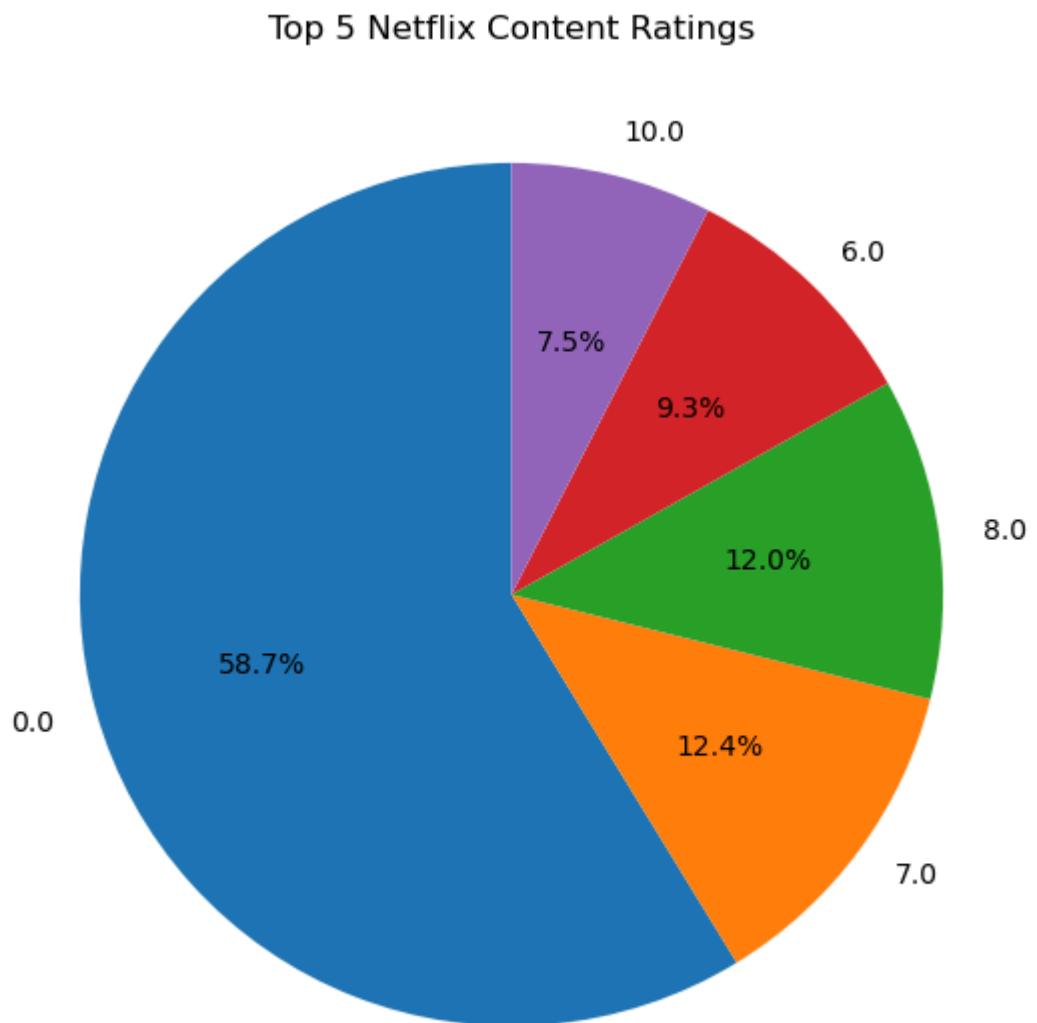
plt.show()
```



```
In [9]: #Pie Chart (Top 5 Ratings)
rating = df['rating'].value_counts().head(5)

plt.figure(figsize=(7,7))
plt.pie(rating.values,
        labels=rating.index,
        autopct='%1.1f%%',
        startangle=90)

plt.title("Top 5 Netflix Content Ratings")
plt.show()
```

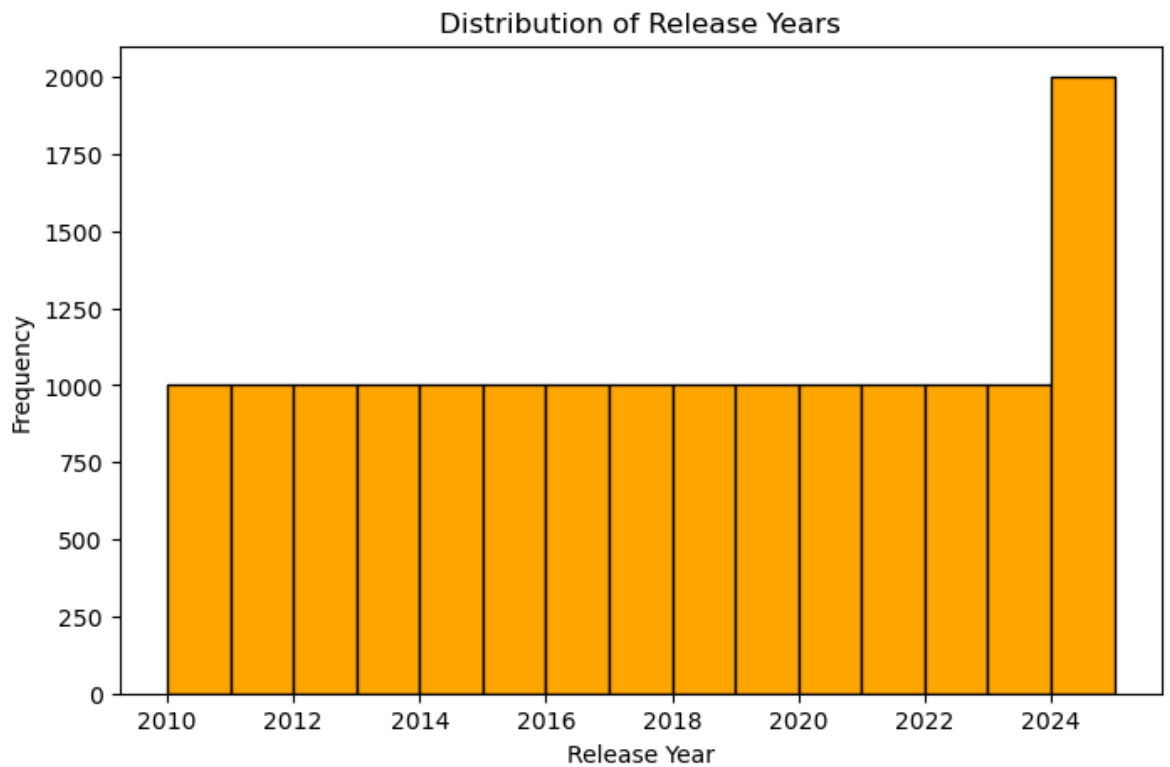


```
In [10]: #Histogram (Release Year Distribution)
plt.figure(figsize=(8,5))

plt.hist(df['release_year'],
         bins=15,
         color='orange',
         edgecolor='black')

plt.title("Distribution of Release Years")
plt.xlabel("Release Year")
plt.ylabel("Frequency")

plt.show()
```



```
In [14]: # Scatter Plot (Popularity vs Vote Average)
```

```
plt.figure(figsize=(8,5))
```

```
plt.scatter(df['popularity'], df['vote_average'], color='red')
```

```
plt.title("Popularity vs Vote Average")
```

```
plt.xlabel("Popularity")
```

```
plt.ylabel("Vote Average")
```

```
plt.grid(True)
```

```
plt.show()
```

```
# Scatter Plot (Popularity vs Vote Count)
```

```
plt.figure(figsize=(8,5))
```

```
plt.scatter(df['popularity'], df['vote_count'], color='blue')
```

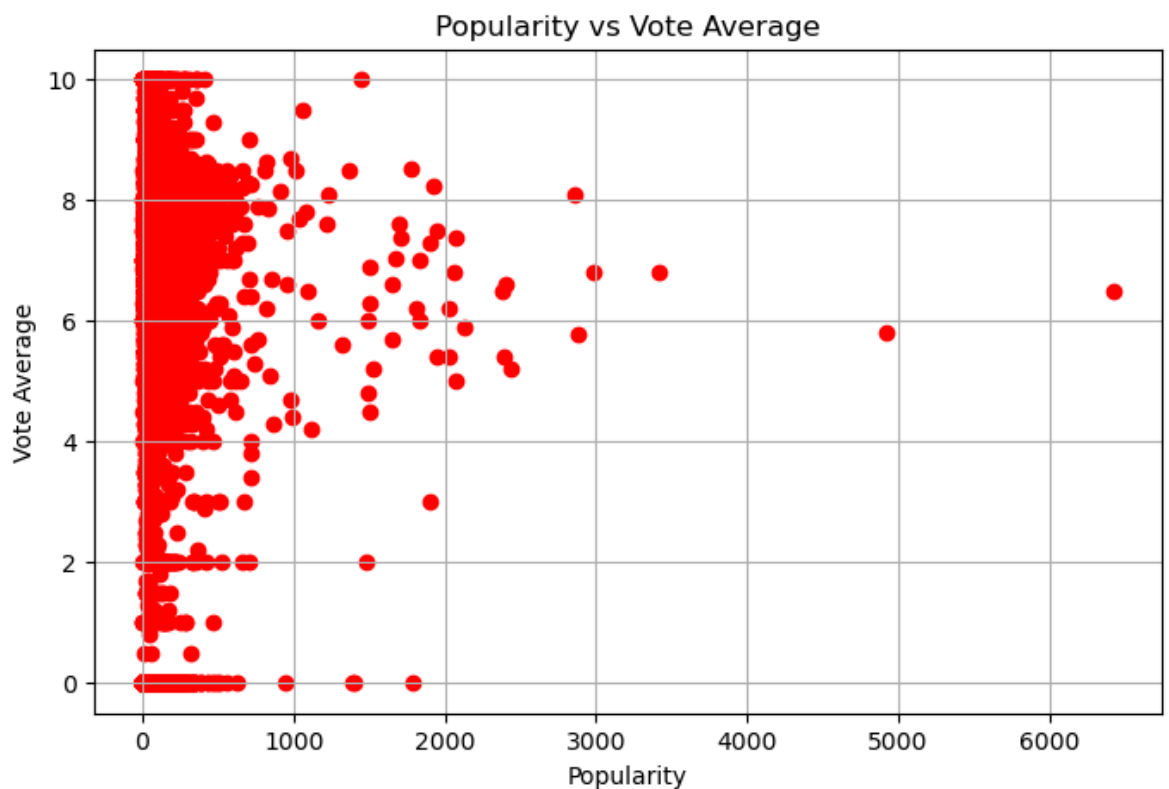
```
plt.title("Popularity vs Vote Count")
```

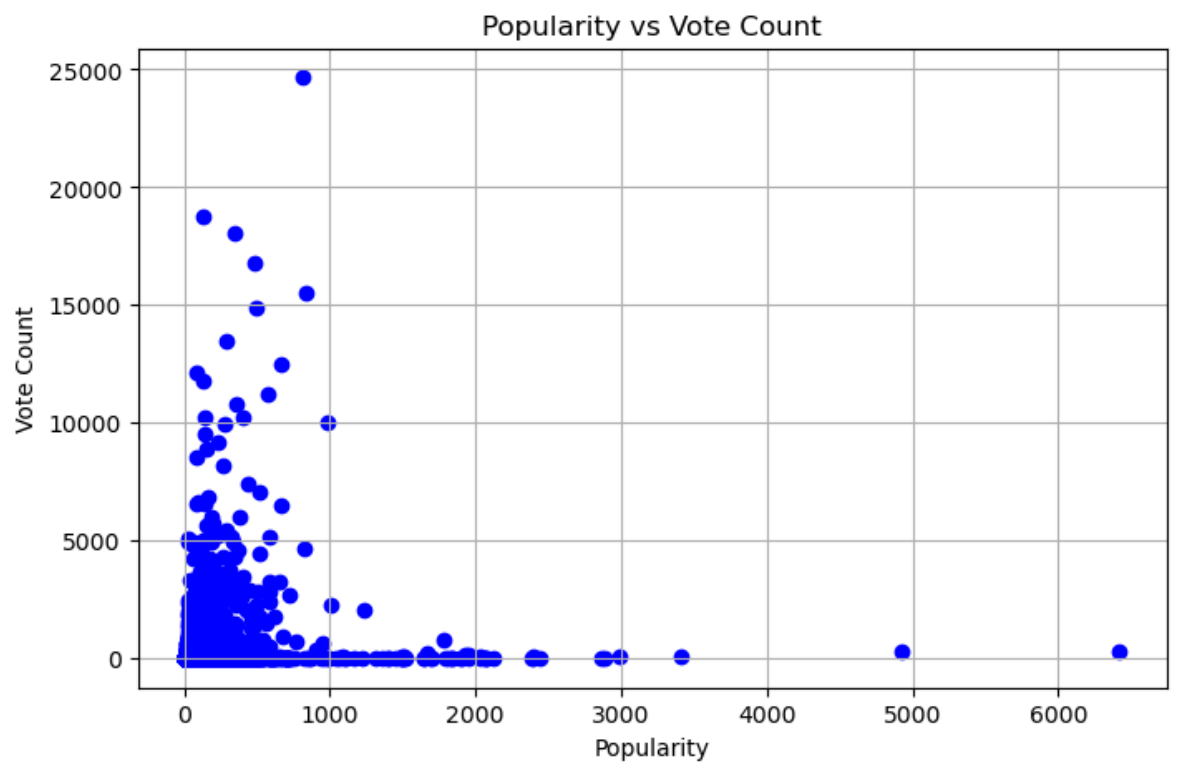
```
plt.xlabel("Popularity")
```

```
plt.ylabel("Vote Count")
```

```
plt.grid(True)
```

```
plt.show()
```





In []: